

Ethanol Production Process Simulation Using Aspen HYSYS: A Multi-Column Distillation and Absorption Approach

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Background & Description:

This project simulates an **ethanol purification process** derived from fermentation using **Aspen HYSYS**, following the flowsheet structure described in the *Ethanol Plant* case study .

Fermentation produces ethanol, water, and trace by-products (methanol, propanols, butanols, pentanols, acetic acid, glycerol, and CO₂). Because ethanol and water form an azeotrope at atmospheric pressure, the separation requires a train of **absorbers** and **distillation columns**.

The main operations included in the simulation are:

1. **CO₂ Vent Separator** – splits CO₂ and liquid fermentation broth.
2. **CO₂ Wash Tower** – absorbs residual ethanol from the CO₂-rich stream.
3. **Concentrator Column** – modeled as an absorber with a side vapor draw to enrich ethanol.
4. **Light Purification Tower** – removes CO₂ and light alcohols.
5. **Rectifier Column** – produces an azeotropic ethanol–water mixture and withdraws fusel oils at an appropriate stage to prevent liquid–liquid phase formation.

Thermodynamic package: The **NRTL property package** with **UNIFAC VLE estimation** is used, as recommended in the tutorial, to describe strongly non-ideal alcohol–water mixtures and fusel oil interactions (page C1-5) .

Observation

During the simulation, **trace impurities could not be included** because recent versions of Aspen HYSYS do not allow entry of mole fractions **below 0.0001**.

Since the fermentation by-products listed in the reference case appear at concentrations lower than this threshold, the software prevented their inclusion.

As a result, the simulation was conducted using a simplified component list (ethanol, water, CO₂), which still allows correct representation of the global separation behavior but limits detailed impurity tracking.

Result Comparison

After comparison, the obtained results are **sensibly similar and of the same order of magnitude** as those from the **standard Aspen HYSYS base simulation model** (C-2.hsc example), despite the absence of ultra-dilute impurities.

This confirms that the core flowsheet — absorbers, concentrator, lights column, and rectifier — behaves consistently with the reference industrial case.

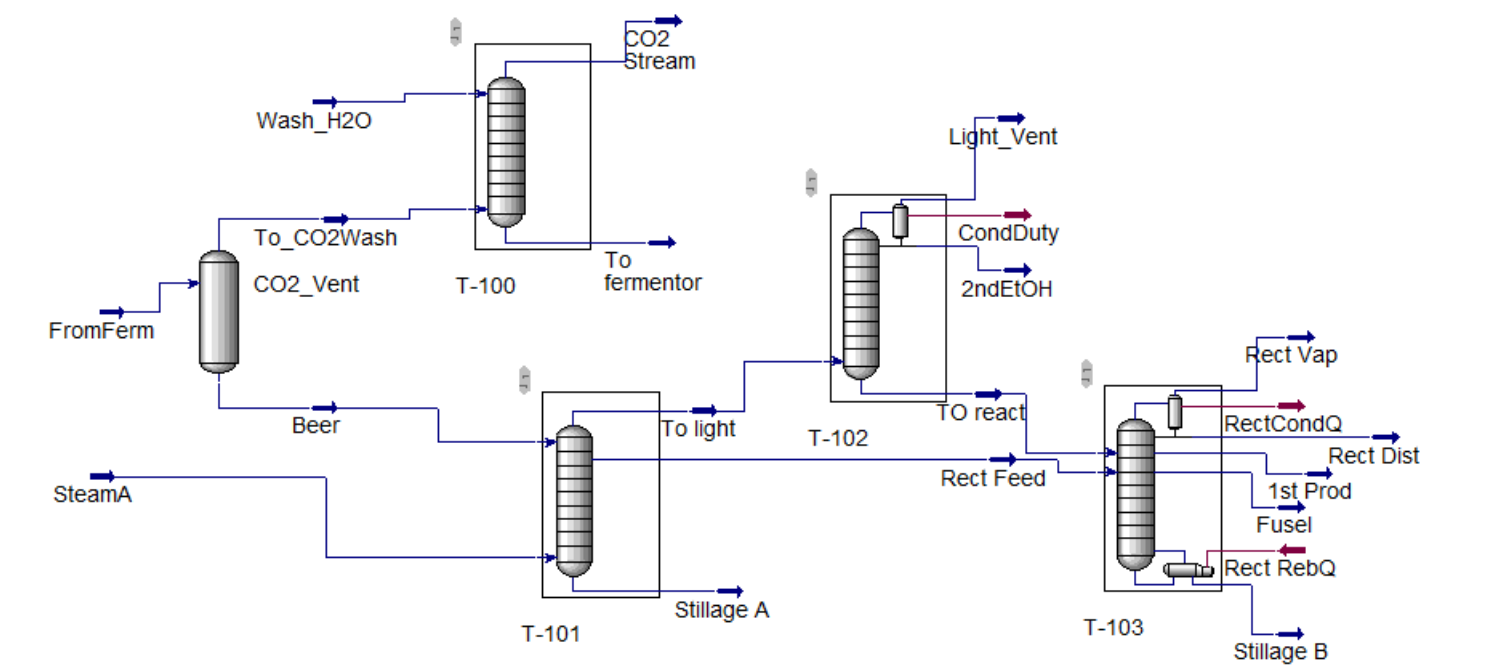
Conclusion

The Aspen HYSYS simulation demonstrates that:

- The ethanol purification sequence can be accurately modeled using multiple absorbers and distillation columns.
- Azeotropic limitations are correctly handled through staged vapor–liquid separation and side draws.
- Even with software limitations regarding extremely small mole fractions, the main ethanol–water–CO₂ system reproduces the expected behavior.
- The results remain comparable in magnitude and trends to the reference HYSYS model, validating the implemented approach.

This methodology can be applied to similar bioethanol purification systems requiring multi-step separation.

Flowsheet



Results:

Master Property Tableau								
Object	Wash_H2O	FromFerm	SteamA	CO2Stream	To fermentor	Light_Vent	ReatVap	StillageB
T (°C)	25.00	30.00	140.00	26.06	33.34	46.45	78.04	99.99
P (kPa)	101.3	101.3	101.3	101.3	101.3	101.3	101.3	101.3
Molar Flow (Kgmole/h)	130	2400	610.6	64.74	131.8	1.60	0.1	254.2
Mole Fract (Ethanol)	0	0.0269	0	<0	0.0086	0.1891	0.8901	0
Mole Fract (H2O)	1	0.9464	1	0.0333	0.9910	0.0525	0.1077	1
Mole Fract (CO2)	0	0.0266	0	0.9667	0.0005	0.7584	0.0022	0

Nb : Insignificant quantities (<0) of components in the product streams can be considered negligible, effectively approximating them to zero without impacting the overall process results or conclusions (see flowsheet).

References: Aspen Hysys Tutorials